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REPLAY

Ticking Bomb: Design Document

Oct 14, 2026 · @David Olamide

Fardoush Hassan & David Olamide · REPLAY Delivery A1 · Hardware party game for events and festivals

1. Short description

> Ticking Bomb is a loud, physical party game for festivals. A ball-shaped bomb ticks down while players race to solve quick puzzles on it: a sum, a color sequence, a dial, a hard shake. Solve it to score or survive. Stuck? Gamble on a pass and dump the bomb on someone else, but pick the wrong button and it blows up in your hands.

330 characters.

2. Player interaction

The bomb is one shared object, so every interaction runs through it. Players touch it, pass it, shout at each other over it, and watch whoever is holding it.

- **With the bomb:** players press colored buttons, turn a knob, and shake the ball to solve puzzles. The bomb answers with ticking, lights, vibration and, eventually, an explosion
- **With teammates:** the holder is rarely alone. Teammates crowd in and shout answers, because a solve scores for the whole team
- **With opponents:** opponents shout wrong answers, and a safe pass dumps the bomb in their hands with whatever time is left
- **With the crowd:** the countdown, the colored callouts and the explosion are loud and visible, so spectators follow along and react without needing the rules explained

Tagging is the hook that pulls everyone in. Each player wears a headband in their own color. When the bomb calls a color, on its screen, its lights and its voice, that player has to grab it. Nobody can drift to the edge of the game, because their color might be next.

3. Playful approach

The game is built around one feeling: **panic you can laugh at.** Hot potato and bomb-defusal are familiar, so players understand the stakes in seconds. We keep every decision fast and every consequence loud.

- **Panic, not paralysis.** Puzzles are short and simple on purpose. A sum you'd solve instantly at a desk becomes hard with a ticking bomb in your hands and five people shouting
- **The gamble.** Passing is always available, but some buttons are secretly dark. Players choose between a slow, safe solve and a fast, risky dump. That choice is where most of the laughter comes from
- **Rising tension.** Every wrong answer cuts the timer, and each round starts a little shorter. The bomb gets more dangerous the longer it's in play
- **The bomb has a personality.** It mocks players, counts down faster when time is short, and punishes failure with a full explosion of sound, light and vibration. Losing is part of the show
- **No reading required.** Rules fit in one breath, and the screen, lights and voice teach the rest while you play

4. The object and the play

In Ticking Bomb the object isn't a controller for a game happening somewhere else. The ball *is* the game. Its rules, timer, puzzles and voice all live inside it, and everything players feel comes from holding it.

The object shapes the experience

- **Weight and presence.** A screen game puts the danger behind glass. Here the danger is in your hands, buzzing against your palms. Holding the ball is the tension
- **Its shape invites passing.** A ball asks to be handed on. That simple physical urge becomes the core decision: solve it, or get rid of it
- **Physical buttons over a touchscreen.** We chose real buttons for the pass gamble because a press you can feel, with no undo, raises the pressure more than a tap on glass
- **The bomb is the referee.** Because it keeps time, checks answers and announces results, nobody argues with each other about the rules. Players argue with the bomb

Performative play

The holder of the bomb is always on stage. Everyone watches them fumble the knob, misread a sum or hesitate over a pass button. That performance is part of the fun, for players and spectators alike.

- **The holder performs panic.** Their choices are public and their mistakes are loud
- **Teammates perform as a chorus,** shouting answers and encouragement
- **Opponents perform as hecklers,** shouting wrong answers on purpose
- **The bomb performs too.** Its mockery lines and over-the-top explosion turn a loss into a moment people laugh at, not a failure people hide from

This is why the game suits a festival floor. It reads well from a distance, and the crowd becomes part of the play even before they join.

Player experience

We want players to walk away laughing and wanting another round. The design supports that in three ways:

1. **Short cycles.** Rounds last seconds to a couple of minutes, so a loss never stings for long
2. **Shared blame, shared glory.** In Teams and Solo, success and failure belong to the group. In Individual mode the stakes are personal, but everyone is in the same panic
3. **Readable feedback.** Every event shows on at least two channels (sound, light, vibration, screen), so players always know what just happened, even in a loud room

Open question for playtesting: is the bomb's personality enough to replace headbands that light up? Right now the headbands only show each player's color, and the bomb does all the calling. If players miss their callout, the headbands may need to react too.

5. The game space

Ticking Bomb is played in the open, at events and festivals like Amaze or GDC. The booth is built around one idea: the game should be as fun to watch as it is to play.

The bomb sits in an open play circle with the crowd all around it

Top-down view of an event booth. Not to scale.

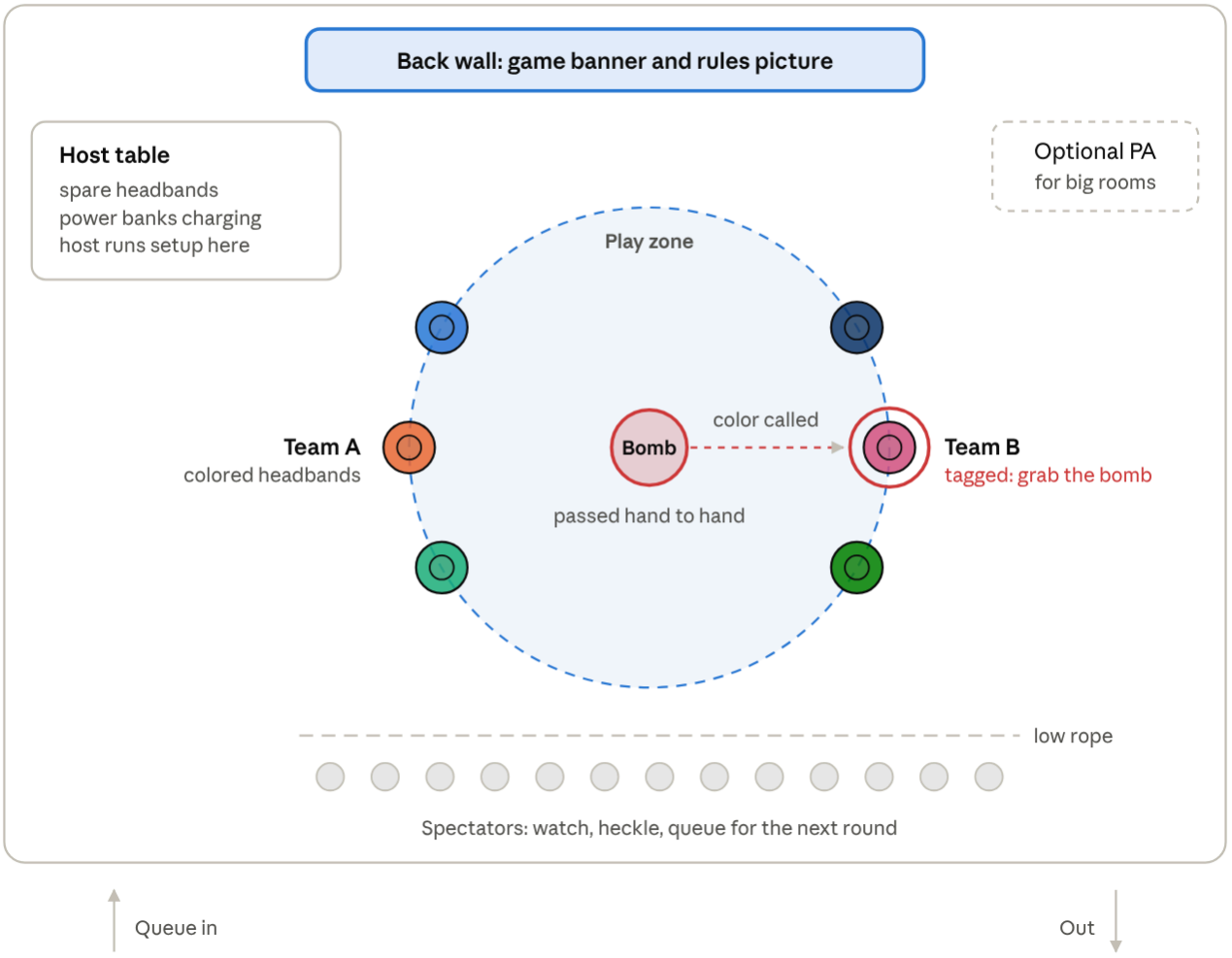


Figure 3 · The game space: an event booth, top-down

- **Play zone:** an open circle with the bomb in the middle and players standing around it in their headbands. In Teams, each team takes one side so spectators can tell them apart
- **Back wall:** a banner and the rules as one simple picture, readable from the queue
- **Host table:** one host runs setup and calibration from the ball, swaps power banks and hands out headbands
- **Spectators:** a low rope keeps the play zone clear while the crowd stays close enough to heckle. Queuing players watch a round before they play, which teaches them the rules for free
- **Sound:** the bomb's own speaker covers small rooms. A PA can be added on a loud show floor
- **Flow:** players queue in on one side and leave on the other, so the line keeps moving between short rounds

The space works the same indoors or outdoors and needs no screen, laptop or Wi-Fi: just the ball, the headbands and room to stand in a circle.

6. Game modes

There are four modes, each tuned for a different kind of crowd. Solo needs no headbands and is the quickest way in.

	Solo (Rapid Fire)	Teams	Individual	Imposter
Players	Everyone around one ball	2 to 8, split into two teams	Each player for themselves	Group with one hidden imposter
Headbands	None	Yes	Yes	Yes
Pass or skip	Skip a puzzle, with dark button risk	Pass to the other team, any time	Pass to another player, any time	Ball passed round for voting
Scoring	Solve the target number to defuse	+1 per solve, -1 per explosion (never below 0)	Explode and you're out	Vote out the imposter
Win	Group defuses the bomb	First team to the points target wins the set	Last player standing	Civilians find the imposter
Best for	Demos, short lines, first-timers	Competitive groups, spectators	Longer sessions, friend groups	Groups who stay a while
Status	Designed	Designed	Proposed	Future idea

6.1 Solo (Rapid Fire)

Everyone plays together on one ball. Solve the target number of puzzles before the timer runs out. Our starting point is 10 puzzles in 5 minutes, both tunable.

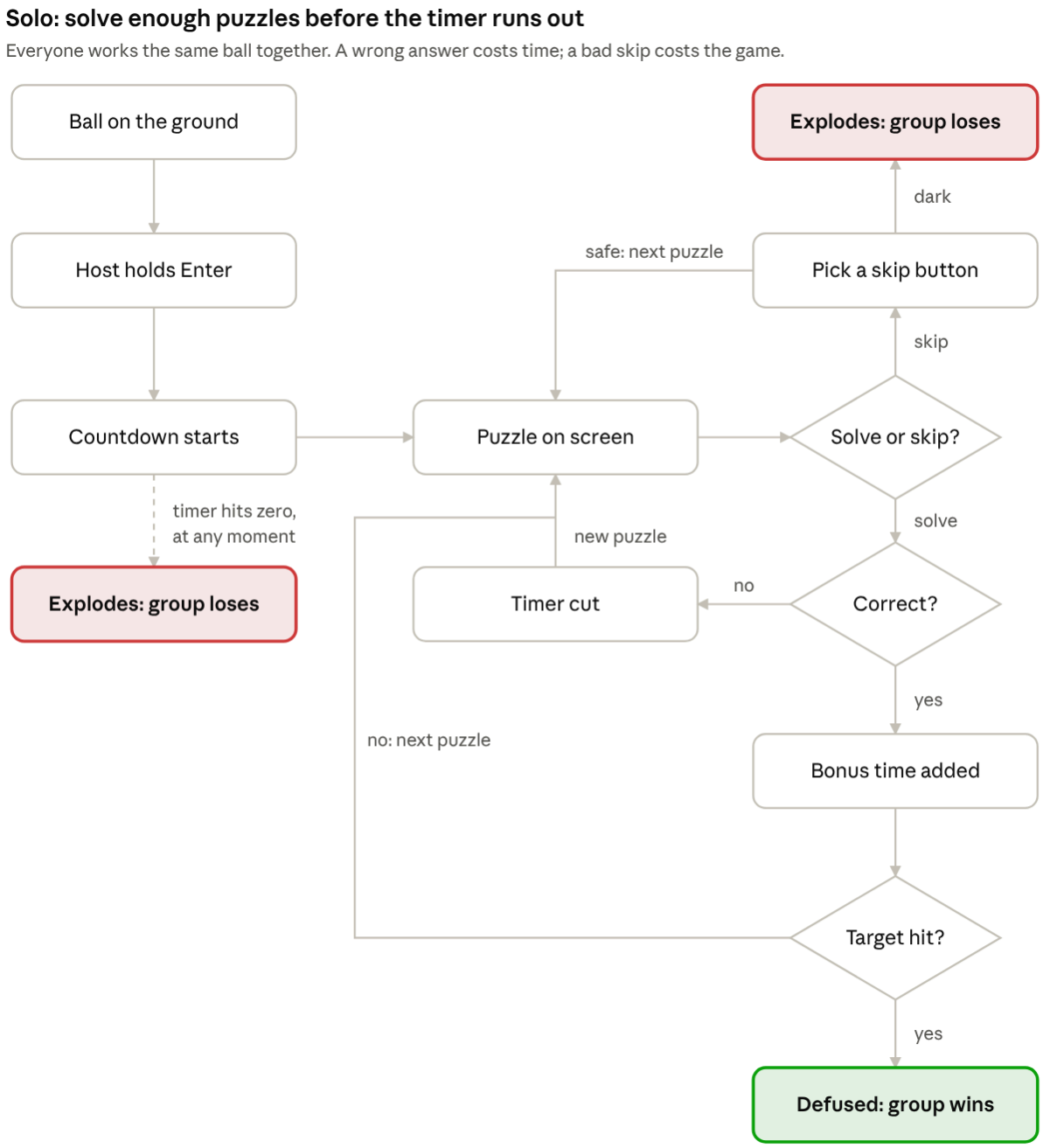


Figure 1 · Solo mode flow

- **Correct:** bonus time is added and the counter goes up
- **Wrong:** the timer is cut and a new puzzle appears
- **Skip:** pick a pass button instead of solving. A safe button skips to the next puzzle. A dark button explodes the bomb and the group loses

6.2 Teams

Two teams take turns with the bomb and score by solving. Passing dumps the bomb on the other team, but risks blowing it up in your hands.

Teams: solve to score, pass to dump the bomb and risk a dark button

Teams take turns. First team to the points target wins the set.

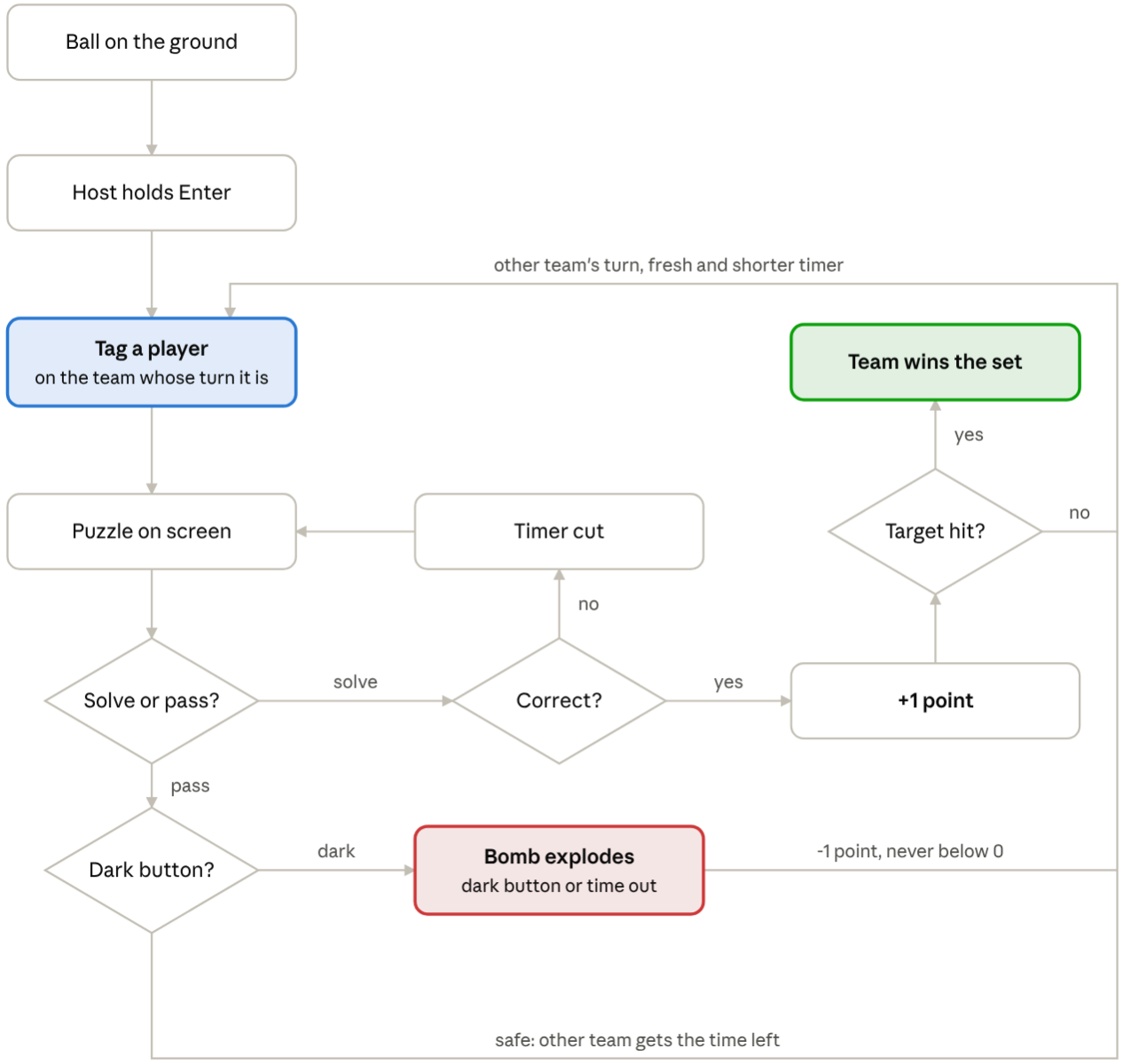


Figure 2 · Teams mode flow

1. Each round, the bomb tags a player on the team whose turn it is, and a fresh timer starts, a little shorter each round
2. The tagged player can **solve** or **pass** at any time, even without trying the puzzle first
3. **Correct solve:** +1 point. The next round goes to the other team
4. **Wrong answer:** the timer is cut. Try again, or pass
5. **Pass:** pick one of the 4 physical buttons. Safe means the other team takes the bomb with the time left. Dark means it explodes on the passer
6. **Explosion:** that team loses 1 point, never below 0. The next round goes to the other team

7. **First team to the points target wins the set**

Passing is a real trade-off. Pass early and you hand the other team a healthy timer and a good shot at the point. Pass late and you hand them a nearly dead bomb, but every second you hold on risks the explosion.

6.3 Individual

Every player for themselves. Explode and you're out, and the last player standing wins. It runs as one elimination game with no sets, so it lasts longer than Teams.

- The bomb tags a random player, who must solve or pass
- **Correct solve:** the bomb is tagged to another player with a fresh, shorter timer
- **Wrong answer:** the timer is cut
- **Pass:** safe sends the bomb to another player with the time left. Dark explodes it on the passer

6.4 Imposter (future idea)

One player is secretly the imposter. The ball is passed around and everyone votes on the imposter's color. Vote out an innocent player and the timer halves and they leave. Find the imposter and the civilians win. We'll design this mode after playtesting the others.

7. Core mechanics and puzzles

Four mechanics run every mode: tagging, solving, the pass gamble and the explosion.

7.1 Tagging

The bomb calls a color and the player wearing it must grab the ball. The callout fires on every channel at once: the screen shows the color name, the speaker shouts it and the LED strip turns that color. Tagging is used in Teams, Individual and Imposter.

7.2 Solving

The holder solves the puzzle shown on the bomb's screen and confirms with Enter. What a solve earns depends on the mode: bonus time in Solo, a point in Teams, a safe handover in Individual. A wrong answer always cuts the timer.

7.3 The pass gamble

The holder can give up on a puzzle at any time by picking one of the 4 physical buttons. One or two are secretly dark.

Mode	Safe button	Dark button
Solo (skip)	Next puzzle, no time lost	Bomb explodes, group loses
Teams (pass)	Other team takes the bomb with the time left	Passer's team loses 1 point
Individual (pass)	Another player takes the bomb with the time left	Passer is out

7.4 Explosion

The bomb explodes when the timer hits zero or a dark button is picked. It's a full show: loud explosion audio, the LEDs strobe, strong vibration, and the screen delivers a mockery line.

7.5 Puzzles

Every puzzle is solved on the ball. They fall into two groups: brain puzzles and physical puzzles.

Puzzle	Type	On screen	What the player does	Input
Maths	Brain	A short sum and 4 answers in a row, in the same order as the buttons	Press the button in the right answer's position, then Enter	Buttons
Number sequence	Brain	A sequence with a gap and 4 options	Press the button for the missing number, then Enter	Buttons
Color sequence	Brain	"Watch the lights" while the LED strip flashes a color sequence	Repeat it on the colored buttons, then Enter	Buttons
Knob to position	Physical	A needle and a target, starting in a random spot	Turn the knob until they line up, then Enter	Rotary knob
Shake the ball	Physical	A meter that fills up	Shake hard until it's full	Accelerometer

Puzzle	Type	On screen	What the player does	Input
Shout at it	Physical	A volume meter	Shout until it fills	Microphone (future idea)

Puzzles are kept short so that the pressure, not the puzzle, is what makes them hard. Difficulty rises through bigger numbers, longer sequences, tighter knob targets and fuller shake meters, and is balanced between teams with some randomness.

8. The bomb

The bomb is self-contained and portable. One ESP32 inside the ball runs the whole game, with no laptop, no base station and no Wi-Fi. Switch it on and it's ready.

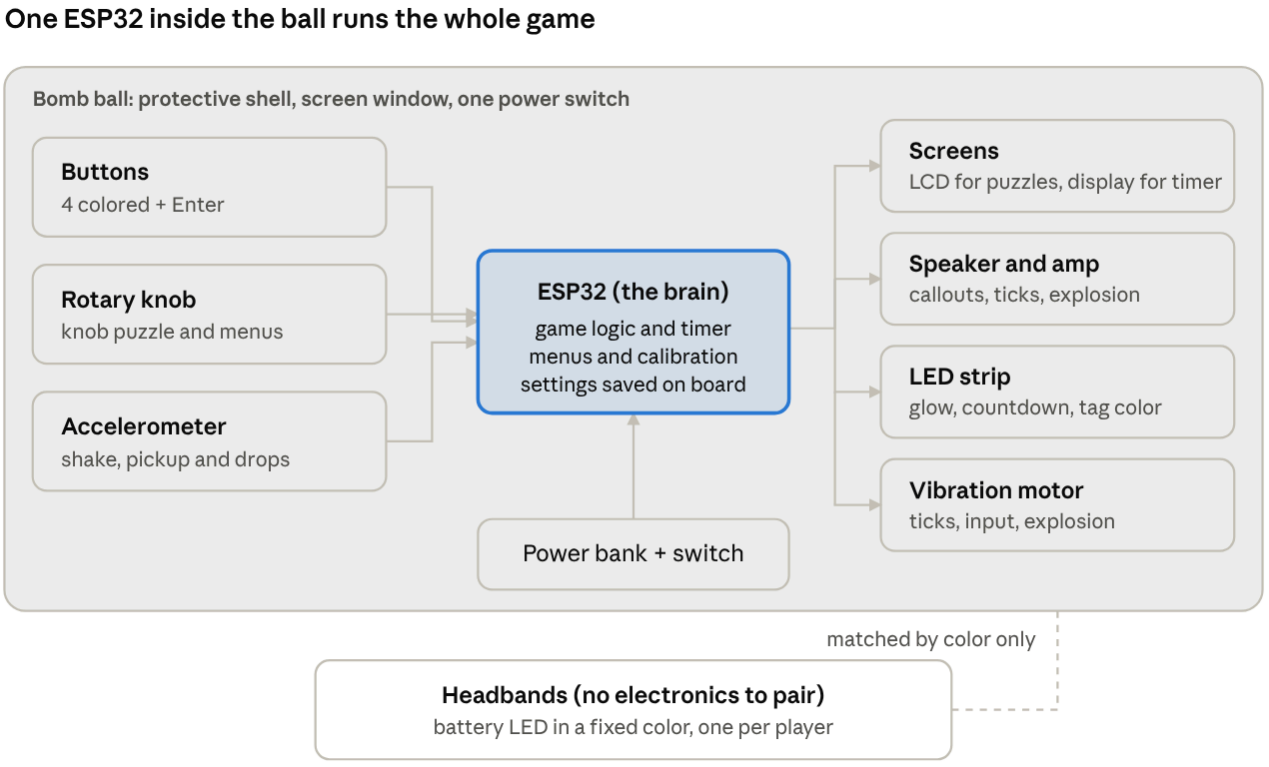


Figure 4 · How the bomb works inside

The headbands are deliberately simple: a battery LED in a fixed color. The bomb does all the signalling, so there's nothing to pair or charge between rounds.

8.1 Feedback channels

Every event shows on at least two channels, so it reads even on a loud festival floor.

Event	Screen	Sound	Lights	Vibration
Tagging	Color name	Shouts the color	Strip turns that color	
Countdown	Timer	Ticking that speeds up	Strip drains	Pulses get faster
Warning (last seconds)	Flashing timer	Alarm beeps	Red flashing	Strong buzzing
Input	Answer highlighted	Click, right or wrong sound		Short buzz
Mockery	Taunt line	Taunt		
Explosion	Boom and a mockery line	Explosion	White strobe	Long, hard buzz

8.2 Parts

Part	Job
ESP32 board	Runs everything
16x2 LCD + alphanumeric display	Puzzles and menus, plus a big countdown
Speaker + PAM8403 amplifier	Callouts, ticking, explosion
Addressable LED strip	Glow, countdown, tag color
Vibration motor	Ticks, input feedback, explosion
4 colored buttons + Enter	Puzzle answers, the pass gamble
Rotary encoder knob	Knob puzzle, menu scrolling
Accelerometer	Shake puzzle, pickup and drop detection
Rocker switch + power bank	One switch to power it all
Round shell	Holds and protects everything, with a window for the screens

Some of the parts we're building with:



Figure 5 · Rotary encoders. Spin freely and press like a button: the dial puzzle and menu control.

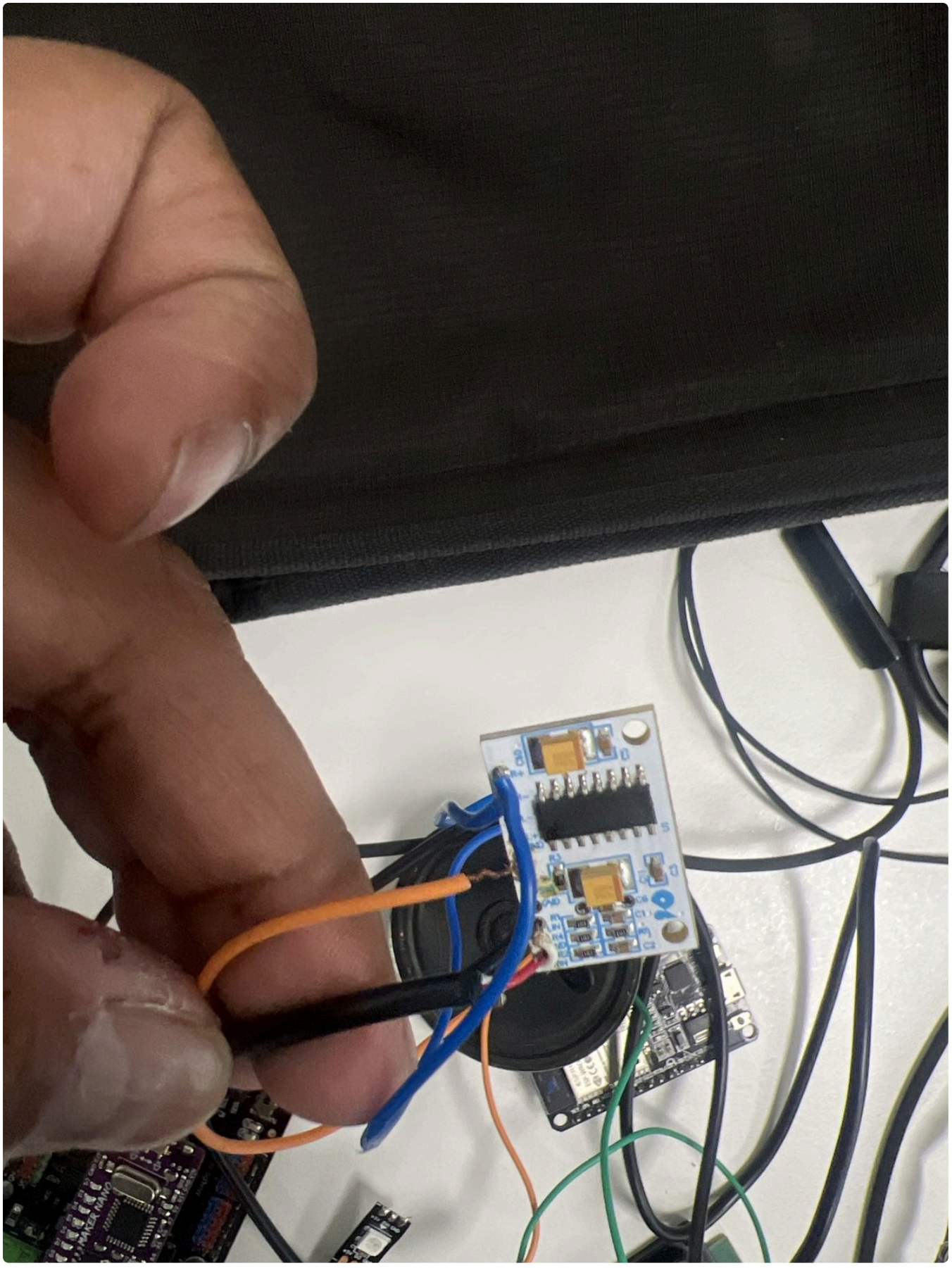


Figure 6 · PAM8403 amplifier wired to the speaker, for the ticking, callouts and explosion.

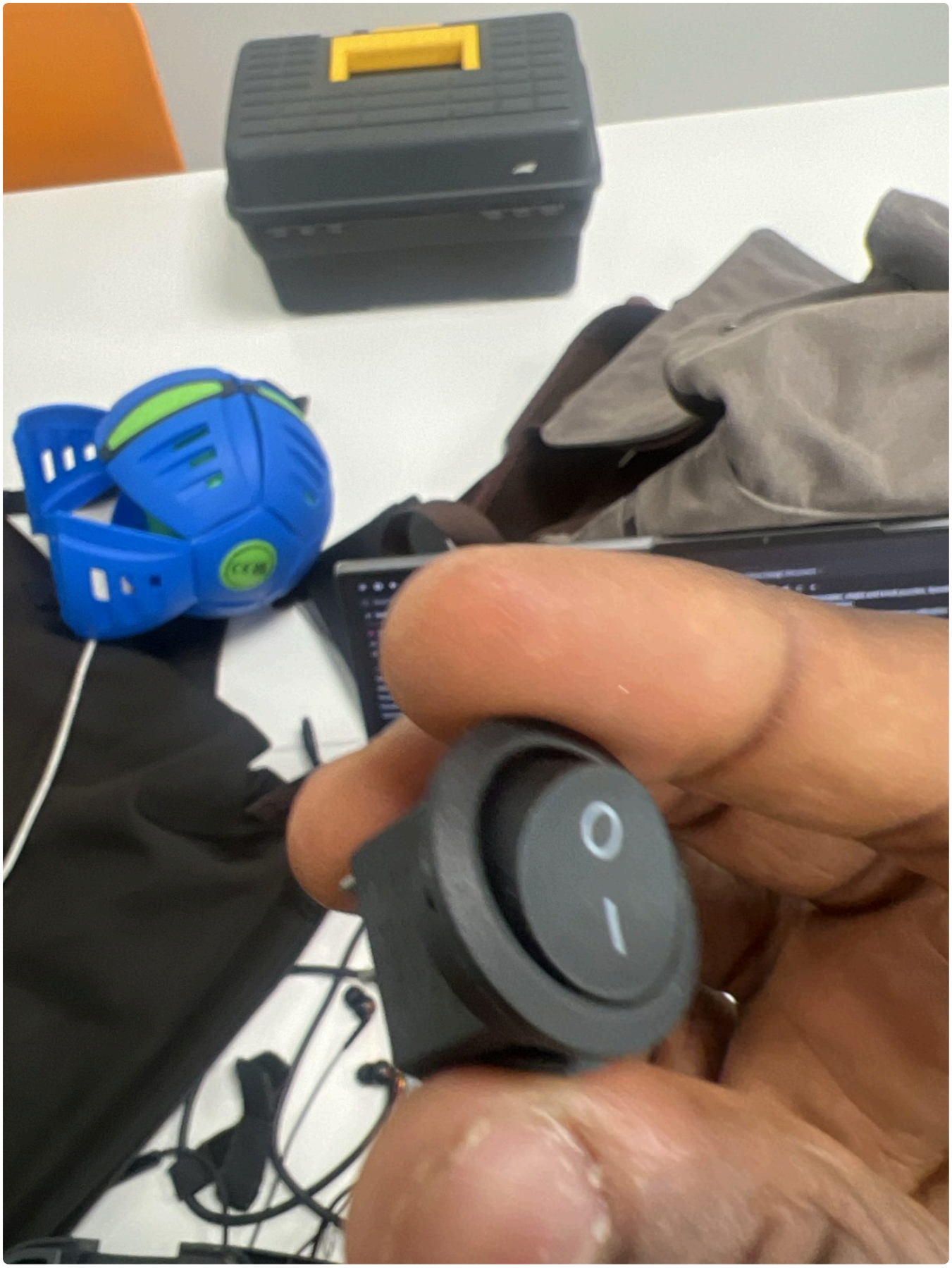


Figure 7 · Round rocker switch. The one switch that turns the bomb on.

9. Setup and calibration

One booth host runs everything from the ball itself, using the knob to scroll and Enter to select. Settings are saved on the ball, so they survive a power-off.

Setup (under a minute)

1. Switch on the ball. It boots to the main menu and shows battery level
2. Pick a mode: Solo, Teams or Individual
3. Teams and Individual: choose the headband colors in play (and the team split for Teams)
4. Run the mode check below
5. Set the ball on the ground and hold Enter to arm it

Hardware check (once a day, or after moving venue)

Check	What the host does	What it confirms
Buttons	Press each one	Every button registers
Knob	Turn a few clicks each way, then press	Direction and press both work
Shake	Hold still, pass normally, then shake hard	Shake triggers on a real shake, not on normal handling
Pickup	Set down, then lift	Ground and pickup detection
Screens, lights, vibration	Watch and feel the test patterns	Every output works
Volume	Play a test callout and explosion	Loud enough for the room

Per-mode checks

- **Solo:** pick a difficulty preset (countdown length, puzzles to defuse, bonus time) and the puzzle mix, then run one practice puzzle of each type
- **Teams:** assign colors to teams, then the bomb calls every color so each player confirms they can hear theirs. Set pass risk (1 or 2 dark buttons of 4), timer and points target
- **Individual:** same as Teams without the team split

Timing, puzzle count, pass risk and the points target are all settings, so we can tune the game between playtests without changing code.

10. Process: sketches and prototype

The game started as hand-drawn flows in a notebook, then moved to a cardboard button panel on the bench.

10.1 Design sketches

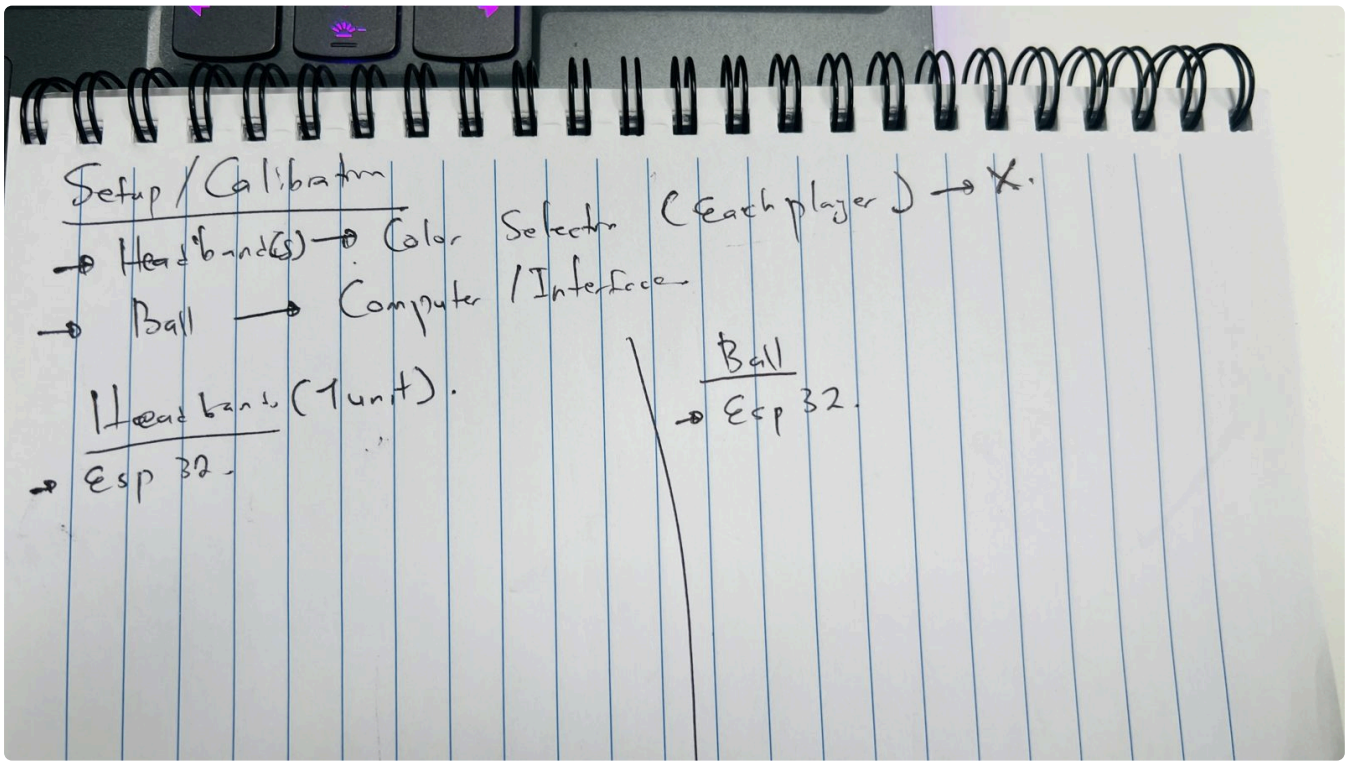


Figure 8 · Setup and calibration. Our first split between headbands and ball, each with its own ESP32. We later cut this to one ESP32 inside the ball.

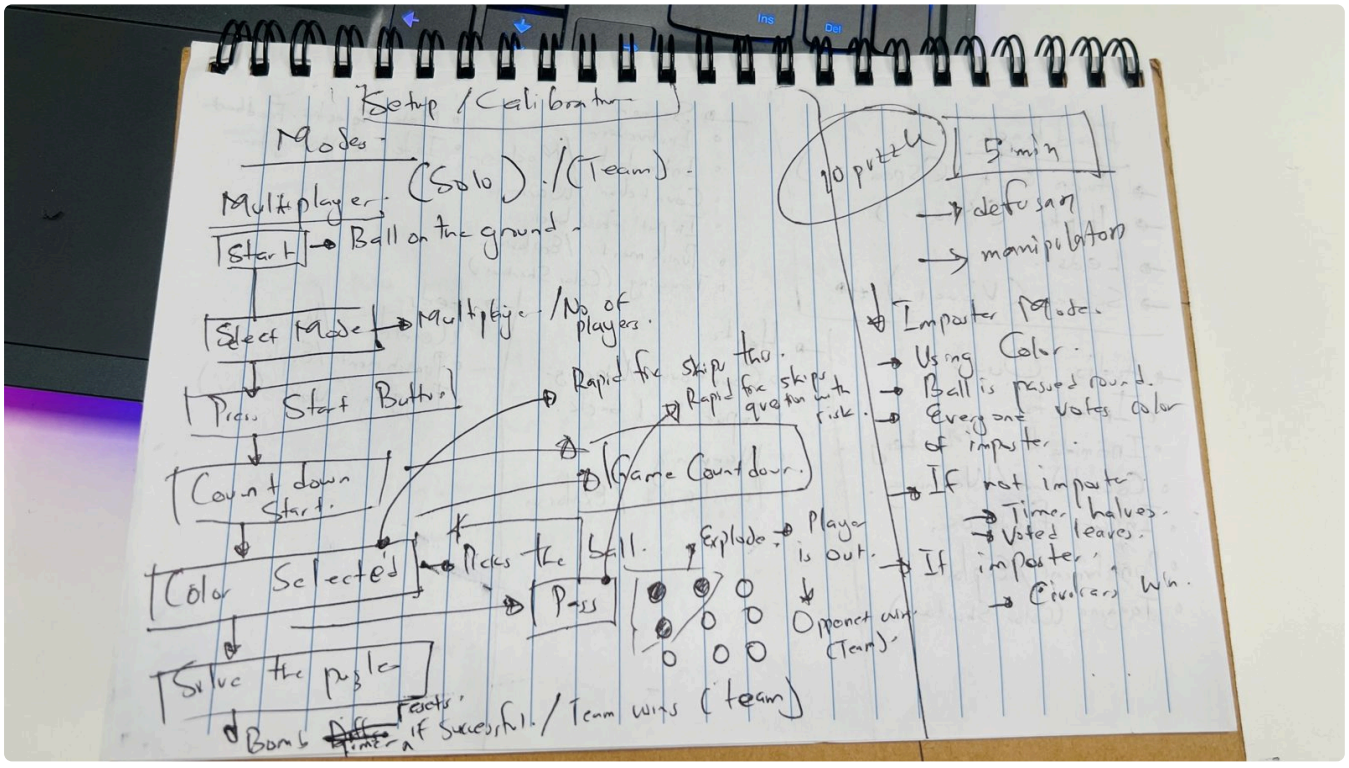


Figure 9 · Game flow. Mode selection, the countdown, color selection, solving, the pass grid with dark buttons, and the first notes on Imposter mode.

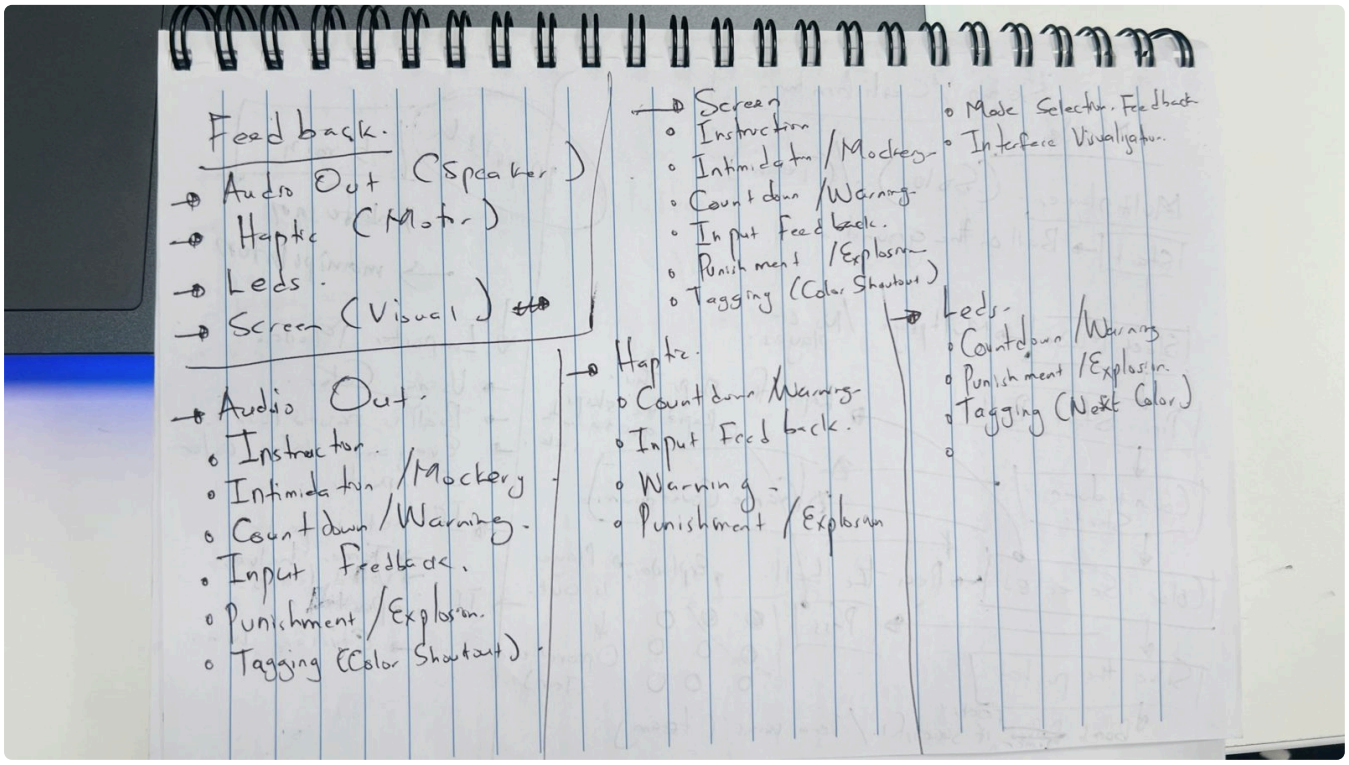


Figure 10 · Feedback. Each channel (audio, haptics, LEDs, screen) mapped to the events it reports.

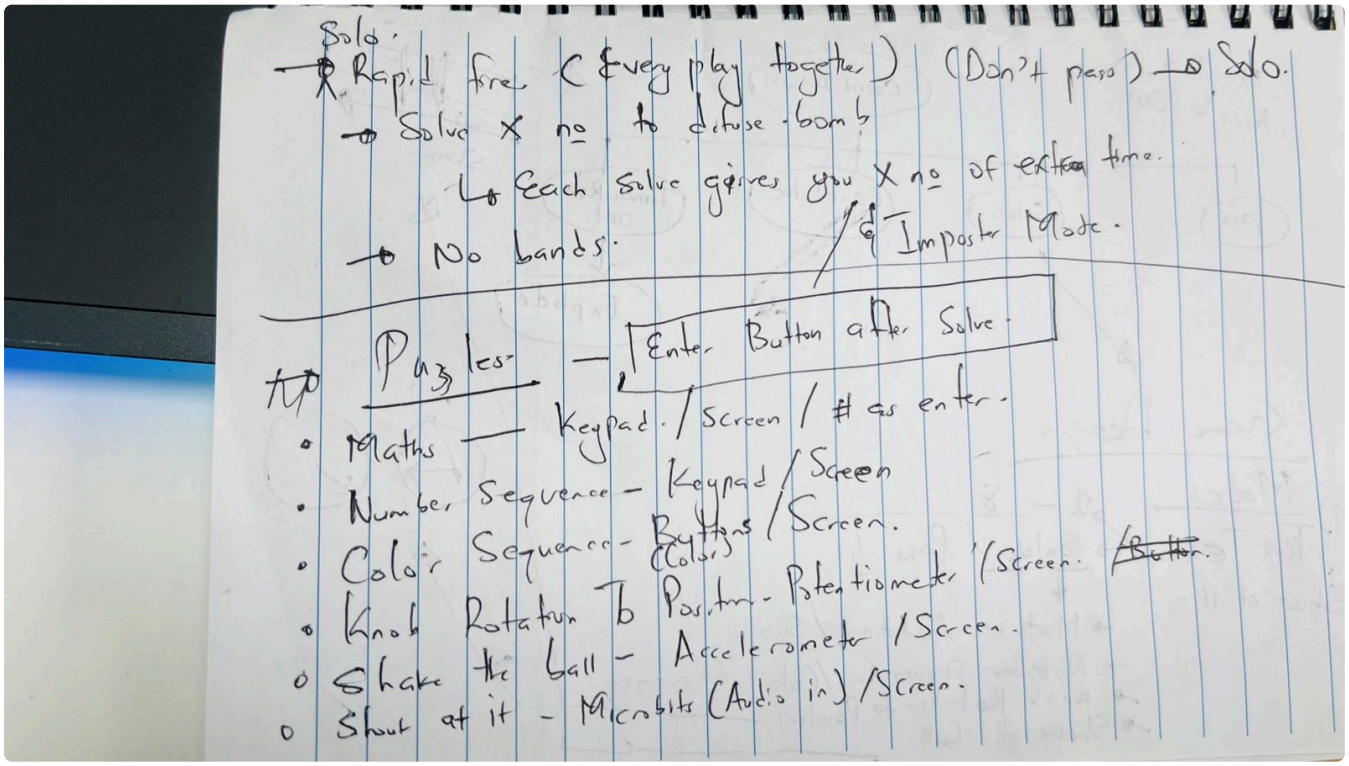


Figure 11 · Solo mode and the puzzle list, with the input each puzzle needs.

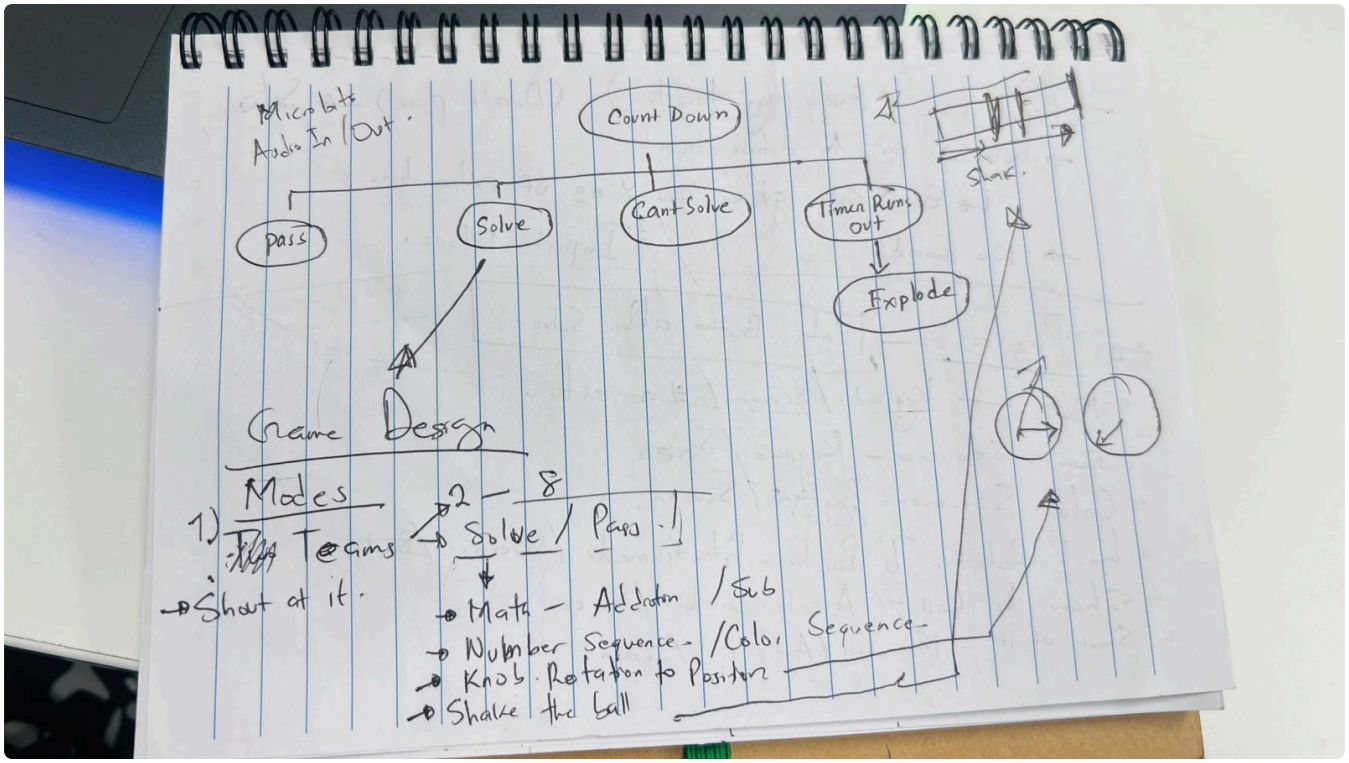


Figure 12 · Game states (countdown, solve, can't solve, pass, explode) and the Teams outline.

10.2 Prototype

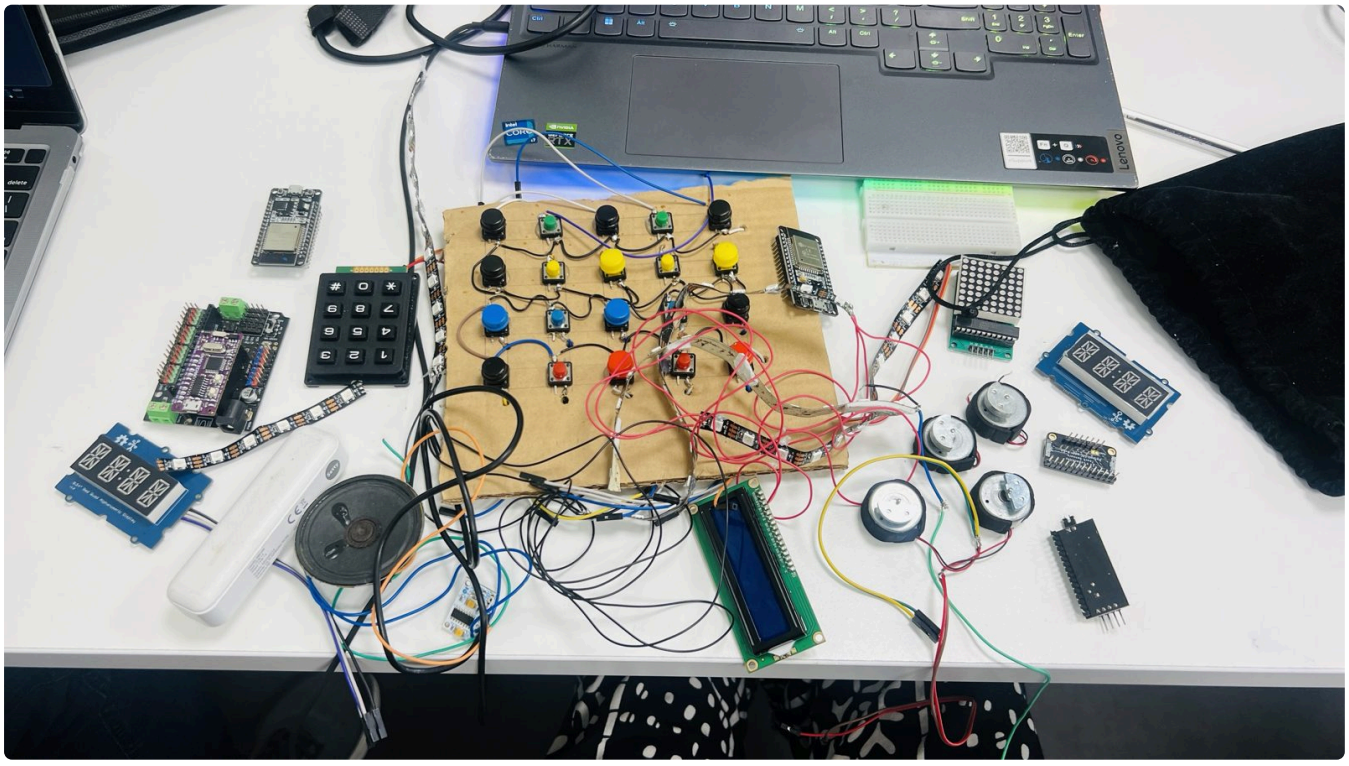


Figure 13 · The bench prototype: a cardboard panel of colored buttons wired to an ESP32, with the LCD, alphanumeric displays, LED strip, vibration motors and speaker around it.

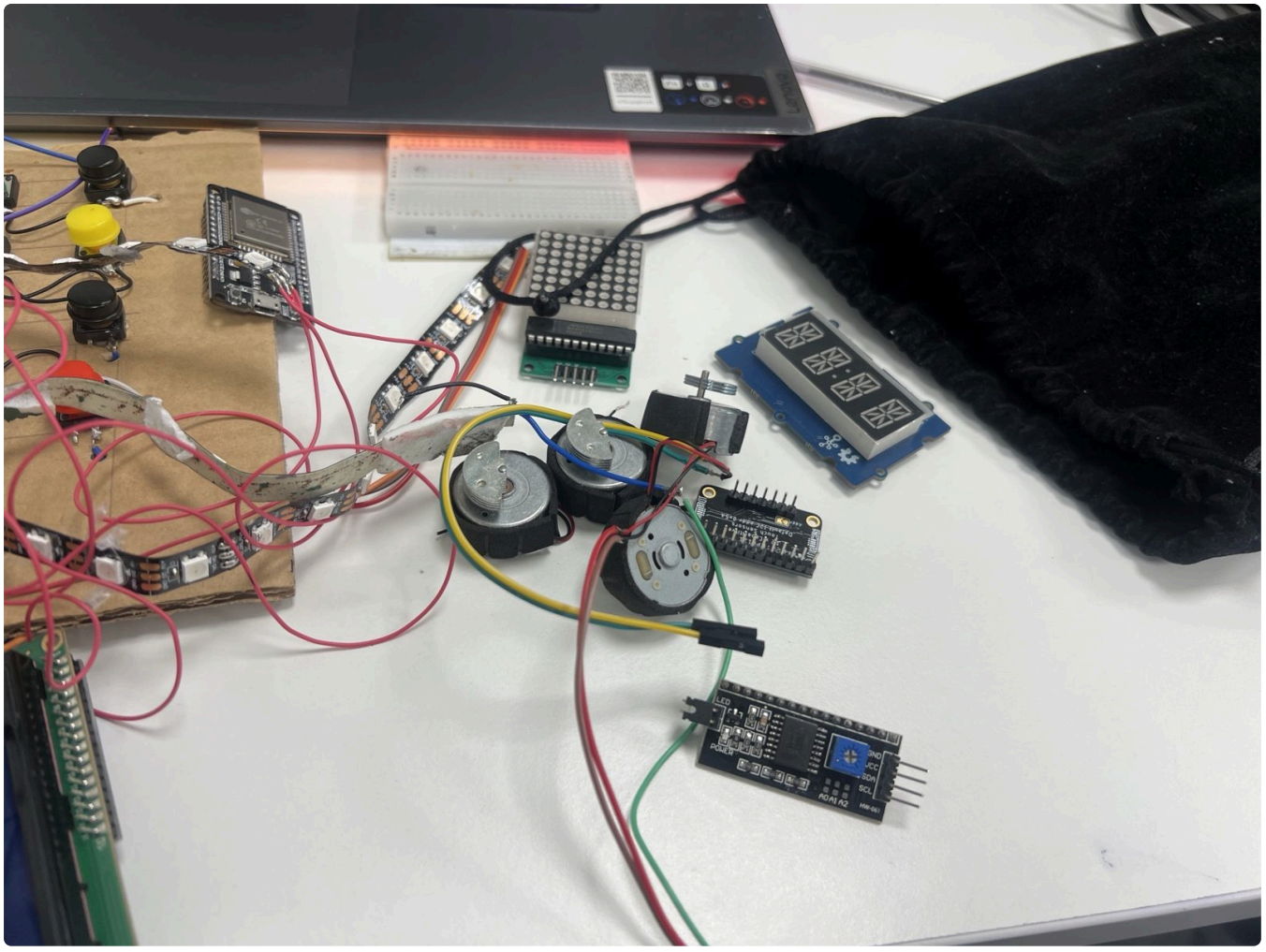


Figure 14 · Close-up of the ESP32, the LED strip, vibration motors, an alphanumeric display and the LCD's I2C board.

10.3 Parts we explored



Figure 15 · BBC micro:bit V2. It has an accelerometer and a microphone, so we can test shake and drop detection now and try the shout puzzle later. Not part of the main build.



Figure 16 · PIR motion sensor. It senses people moving nearby, not the ball moving, so it can't detect a fall. A possible attract mode: the bomb wakes up and taunts people who walk past.

11. Open questions and next steps

These get settled through prototyping and playtesting.

- **Game name** for event listings, styled so it reads clearly as a toy
- **Screen setup:** LCD for puzzles plus an alphanumeric display for the timer, or something else
- **Screen placement and protection** on the ball: padding, a recessed window or a clear cover
- **Voice callouts:** are they loud enough to replace headbands that light up? To test in a noisy room
- **Individual mode:** after a solve, does the bomb go to a random player or does the solver choose? How many players does it need?
- **Balance:** timer length, how much each wrong answer costs, dark button odds and the points target

Next playtest goals: check that strangers can learn the game in one explanation, that the pass gamble creates real tension, and that the explosion makes people laugh rather than feel cheated.